ModelIt!

Teacher's Guide

Gene Regulatory Networks

From Transcription Factor to Phenotype — Modeling the Logic of Gene Expression

Quick Reference

Grade Level	Grade 09L3
NGSS Standard	HS-LS3-1, HS-LS1-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS3-1, HS-LS1-4

Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring; use a model to illustrate the role of cellular division in producing and maintaining complex organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Asking Questions and Defining Problems** — Students develop a multi-scale gene regulatory model that connects molecular interactions (transcription factor-DNA binding) to cellular outcomes (protein levels, cell fate), asking questions about how regulatory logic creates emergent biological behavior.
Disciplinary Core Idea	**LS3.A Inheritance of Traits / LS1.A Structure and Function** — DNA contains instructions for protein production regulated by transcription factors and epigenetic modifications; cellular function emerges from the coordinated expression of gene regulatory networks.
Crosscutting Concept	**Cause and Effect / Systems and System Models** — Students trace causal chains from transcription factor binding through mRNA and protein dynamics to cellular phenotype, analyzing the gene regulatory network as a complex system with emergent properties including switches, oscillators, and stability.

CAST Testing Alignment

MULTIPLE CHOICE:

A mutation increases the mRNA Degradation Rate of a tumor suppressor gene by 5-fold while leaving Transcription Rate unchanged. Based on your model, what is the most likely consequence for steady-state protein level and cell behavior?

CONSTRUCTED RESPONSE:

Using your gene regulatory network model, explain how two genetically identical cells can have completely different gene expression profiles and become different cell types. Reference at least three model components — including Epigenetic Modification and Feedback Signal Strength — in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson introduces students to gene regulatory networks — the molecular logic circuits that control which genes are expressed in every cell. Biotech skill focus: Network dynamics and feedback circuit analysis. Understanding gene regulation is fundamental to both basic biology (how does a fertilized egg become a complete organism?) and biotechnology (how do we engineer cells to produce desired proteins or behaviors?).

The Central Dogma and Its Regulation

The flow of genetic information from DNA to RNA to protein — the central dogma of molecular biology — is regulated at every step. Transcriptional regulation (which genes are copied into mRNA) is controlled by transcription factors that bind to promoter and enhancer regions of DNA, either recruiting or blocking RNA polymerase. Post-transcriptional regulation controls mRNA processing (splicing, capping, polyadenylation), mRNA stability (degradation by ribonucleases and microRNAs), and mRNA localization. Translational regulation controls how efficiently ribosomes convert mRNA into protein, influenced by initiation factor availability, codon usage, and ribosome density. Post-translational regulation controls protein folding, modification (phosphorylation, ubiquitination, glycosylation), localization, and degradation. Each regulatory layer adds complexity but also adds control points that the cell — or the engineer — can exploit.

Feedback Loops: The Logic of Biological Circuits

Gene regulatory networks use feedback loops to create sophisticated dynamic behaviors from simple molecular interactions. Negative feedback occurs when a gene's protein product represses its own transcription — this creates a self-correcting homeostatic circuit that maintains stable protein levels despite perturbations. If protein level rises above the setpoint, increased repression reduces transcription; if it falls below, reduced repression increases transcription. Positive feedback occurs when a gene's protein product activates its own transcription — this creates a bistable switch with two stable states (high expression or low expression) and a sharp threshold for switching between them. Positive feedback is how cells make irreversible fate decisions during development: once a master regulator crosses the threshold and activates its own gene, the cell locks into a new identity. Combined negative and positive feedback creates oscillators — circuits that rhythmically pulse gene expression, underlying circadian clocks, cell cycle progression, and somite segmentation in embryonic development.

Epigenetic Regulation: Above the Genome

Epigenetic modifications add a layer of gene regulation that operates above the DNA sequence level — modifying gene accessibility without changing the genetic code. DNA methylation adds methyl groups to cytosine bases in CpG dinucleotides, which recruits repressive protein complexes that condense chromatin and silence gene expression. Histone modifications — acetylation (generally activating), methylation (activating or repressing depending on position), phosphorylation, and ubiquitination — alter the physical structure of chromatin, determining whether DNA is accessible to transcription factors and RNA polymerase. These modifications are maintained through cell division by dedicated maintenance enzymes, allowing a liver cell and a neuron to maintain their distinct gene expression profiles through hundreds of cell divisions despite having identical DNA. Epigenetic dysregulation is increasingly recognized as a driver of cancer (global hypomethylation and focal hypermethylation of tumor suppressor genes), aging, and developmental disorders.

Gene Regulatory Networks and Disease

Many diseases result from disruptions in gene regulatory networks rather than mutations in the protein-coding genes themselves. Cancer frequently involves mutations in transcription factors (p53, MYC, RB) that disrupt the regulatory circuits controlling cell division — a single mutation in the p53 gene disrupts the network that senses DNA damage, arrests the cell cycle, and triggers apoptosis, removing a critical safety brake on cell proliferation. Developmental disorders like holoprosencephaly result from mutations in the Sonic Hedgehog (SHH) signaling pathway's regulatory elements — the gene itself is intact, but the regulatory network that controls where and when it's expressed is disrupted. Autoimmune diseases involve dysregulation of the transcription factor FOXP3, which controls regulatory T-cell differentiation — without proper

FOXP3 expression, the immune system attacks the body's own tissues. Understanding gene regulatory network architecture is therefore essential for understanding disease mechanisms and designing therapeutic interventions.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten components spanning signal input (Transcription Factor Concentration), molecular interaction (Promoter Binding Affinity), mRNA dynamics (Transcription Rate, Degradation Rate), translation dynamics (Ribosome Availability, Translation Rate), protein dynamics (Degradation Rate), regulatory feedback (Feedback Signal Strength), epigenetic context (Epigenetic Modification), and system-level outcome (Regulatory Network Stability).

E **Establish Relationships**

Students determine that Transcription Factor Concentration and Epigenetic Modification are the primary external/upstream inputs — one representing the signal and the other representing chromatin accessibility — while all other components emerge as dynamic responses to these inputs and to each other through production-degradation balances and feedback loops.

V **Visualize & Model**

Students build a gene regulatory network model connecting signal input through the transcription-translation cascade to protein output, with feedback loops that create dynamic behaviors (steady state, switching, or oscillation) depending on circuit architecture.

E **Evaluate Outcomes**

Students run steady-state, positive-feedback-switch, and epigenetic-silencing scenarios to discover how the same molecular components create fundamentally different behaviors depending on feedback structure and chromatin state.

R **Revise & Extend**

Students add MicroRNA Regulation, Chromatin Remodeling Activity, or Stochastic Noise Level to explore post-transcriptional regulation, chromatin dynamics, and the role of molecular noise in cell fate decisions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"You have 20,000 genes. A brain cell uses about 8,000 of them. A skin cell uses a different 8,000. Same DNA, completely different cells. Who decides?"*

Watch For

Display an image comparing cell types from the same organism. Ask: If they have the same DNA, why are they different? What controls which genes are active?

Slide 2: Learning Objectives

3 min

Say: *"Today you're building the control system that runs every cell in your body. Transcription factors are the decision-makers, feedback loops are the logic circuits, and epigenetics is the master switch."*

Watch For

Pre-teach transcription factor and feedback loop. Use the analogy: a transcription factor is like a key for a specific lock on a gene — no key, gene stays closed.

Slide 3: Big Question

3 min

Say: *"Here's the paradox: your liver cells and neurons have identical DNA. But a liver cell makes albumin and a neuron makes dopamine. Something BESIDES DNA is deciding which genes to use. What is it?"*

Watch For

Think-pair-share: Students brainstorm mechanisms that could make the same DNA produce different cell types. Collect ideas — they'll verify these against the model.

Slide 4: LEVER Framework

2 min

Say: *"We're modeling the entire pipeline from a regulatory signal arriving at a gene to the final protein product — and the feedback loops that make this pipeline intelligent."*

Watch For

Preview LEVER steps. Emphasize that gene expression is not a one-way street — the output feeds back to change the input, creating circuits with logic.

Slide 5: Activity 1: Components

8 min

Say: *"Ten components. Two are upstream inputs that set the stage. Eight are dynamic responses. But here's the twist — some of those responses loop BACK to become inputs through feedback."*

Watch For

Guide sorting of external (Transcription Factor Concentration, Epigenetic Modification) versus internal components. Discuss how Feedback Signal Strength blurs the line between input and output.

Slide 6: Activity 2: Connections

8 min

Say: *"At every level — mRNA and protein — there's a race between production and destruction. Steady-state levels depend on the ratio. And the feedback loops make this cascade self-aware."*

Watch For

Students trace the cascade: TF binds promoter, mRNA is made AND destroyed, surviving mRNA is translated AND protein is destroyed, protein feeds back. Map positive versus negative feedback consequences.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios. In scenario 1, the gene reaches a stable level. In scenario 2, it switches to an entirely different behavior — same components, different circuit architecture. In scenario 3, nothing works at all. Why?"*

Watch For

Students predict, then simulate each scenario. The positive feedback switch is the key discovery — understanding how the same parts create qualitatively different behaviors depending on circuit wiring.

Slide 8: Discoveries

5 min

Say: *"You just modeled the logic that controls every cell in your body. When this logic breaks — when a feedback loop fails or an epigenetic mark is lost — the result is disease. Cancer, developmental disorders, autoimmunity — they're all regulatory network failures."*

Watch For

Connect each scenario to a disease mechanism. Positive feedback stuck ON = cancer (constitutively active oncogene). Epigenetic silencing of tumor suppressor = cancer. Feedback loop disrupted = autoimmune disease.

Slide 9: STEM Challenge

12 min

Say: *"A water agency needs a cheap biosensor for lead contamination. Design a living gene circuit — using the regulatory components you just modeled — that detects lead in water and glows green. This is real synthetic biology."*

Watch For

Groups design synthetic gene circuits: sensor element (metal-responsive TF), circuit logic (feedback for sensitivity), output (fluorescent protein), and calibration strategy. Present designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Transcription Factor Concentration, Epigenetic Modification
- **INTERNAL:** Promoter Binding Affinity, mRNA Transcription Rate, mRNA Degradation Rate, Ribosome Availability, Protein Translation Rate, Protein Degradation Rate, Feedback Signal Strength, Regulatory Network Stability

Relationships

- **Transcription Factor Concentration to mRNA Transcription Rate:** POSITIVE (+) — Higher Transcription Factor Concentration increases the probability of promoter occupancy, recruiting more RNA polymerase molecules to the gene and increasing the rate of mRNA synthesis.
- **Promoter Binding Affinity to mRNA Transcription Rate:** POSITIVE (+) — Stronger Promoter Binding Affinity means each transcription factor molecule stays bound longer and recruits RNA polymerase more effectively, amplifying the transcriptional response at any given transcription factor concentration.
- **mRNA Degradation Rate to Protein Translation Rate:** NEGATIVE (-) — Faster mRNA degradation reduces the steady-state pool of translatable mRNA molecules, meaning fewer templates are available for ribosomes to produce protein, reducing the overall translation output.
- **Ribosome Availability to Protein Translation Rate:** POSITIVE (+) — More available ribosomes means more mRNA molecules can be simultaneously translated, increasing the rate of protein production from the available mRNA pool.
- **Protein Degradation Rate to Feedback Signal Strength:** NEGATIVE (-) — Faster protein degradation reduces the steady-state protein pool available to generate feedback signals, weakening the feedback loop and reducing the circuit's ability to self-regulate.
- **Epigenetic Modification to mRNA Transcription Rate:** NEGATIVE (-) when silencing — Repressive epigenetic marks (DNA methylation, H3K27 trimethylation) compact chromatin structure and block transcription factor access to the promoter, reducing mRNA Transcription Rate to near zero regardless of Transcription Factor Concentration.

Key Discoveries

- Steady-state protein levels are determined by the RATIO of production to degradation at every level — doubling transcription rate doubles mRNA only if mRNA Degradation Rate stays constant, and doubling mRNA doubles protein only if Ribosome Availability and Protein Degradation Rate remain unchanged
- Positive feedback loops create bistable switches — the network flips between two stable states (high expression and low expression) with a sharp threshold, explaining how genetically identical cells can adopt completely different fates during development
- Epigenetic modifications act as a master override — when chromatin is closed by DNA methylation or repressive histone marks, no amount of Transcription Factor Concentration can activate the gene, explaining why a liver cell and a neuron have identical DNA but completely different gene expression profiles
- Negative feedback loops create stability and noise reduction — the protein represses its own gene, creating a self-correcting circuit that maintains consistent output despite random fluctuations in any individual component

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Genes are either 'on' or 'off' like a light switch"	While some gene regulatory circuits DO behave as switches (positive feedback bistable systems), most gene expression is graded — genes can be expressed at any level from barely detectable to maximum capacity. The model demonstrates that steady-state expression is determined by the balance of production and degradation rates at every level, producing a continuous range of output levels. Even 'switch-like' genes have intermediate states during transitions. The light switch analogy is especially misleading because it implies gene regulation is simple — in reality, a single gene can be regulated by dozens of transcription factors, each binding with different affinity, creating a complex dose-response landscape.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"DNA methylation permanently silences genes forever"	While DNA methylation is relatively stable and heritable through cell division, it is NOT permanent or irreversible. Cells express TET enzymes (TET1, TET2, TET3) that actively demethylate DNA, and passive demethylation occurs when maintenance methyltransferases fail to copy methylation patterns during DNA replication. During embryonic development, there are two waves of global demethylation (in the early embryo and in primordial germ cells) where most methylation marks are erased and re-established. In cancer, aberrant demethylation of silenced oncogenes and aberrant methylation of active tumor suppressors both contribute to disease. Epigenetic therapy drugs (5-azacytidine, decitabine) work by blocking DNA methyltransferases, demonstrating that methylation is reversible.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More transcription factor always means more gene expression"	The relationship between Transcription Factor Concentration and gene expression is NOT linear — it follows a sigmoidal (S-shaped) saturation curve. At low TF concentrations, promoter occupancy is minimal and expression is low. At intermediate concentrations, small changes in TF produce large changes in expression (the sensitive range). At high concentrations, the promoter is fully occupied and adding more TF has no effect (saturation). Furthermore, if Epigenetic Modification has closed the chromatin, no amount of TF will activate the gene. And in circuits with negative feedback, increasing TF concentration is partially counteracted by the feedback loop, producing a smaller net increase in expression than expected.